Esko Ilmari Vahteristo
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SUMMARY

I build perception systems that work as products, not just on paper: cameras, LiDAR, radar and lasers on edge hardware, in factories and heavy industry. At a seven-person startup I worked directly with customers and built four systems that are now deployed. Along my work achievements, I've published academic papers, won competitions, and developed open-source software.
Sensors: Lidar, radar, laser triangulation, cameras, IMU
Skills: ML, CV, optimization, edge compute, PLC, AI and agentic work, super- and parallel computing
Software: python, ONNX, torch, tensorflow, OpenCV, C, git, docker

WORK EXPERIENCE

Software engineer (ML/CV), Virelabs, 05/2025 - 8/2026

- Developed software for edge devices in industrial environments utilizing cameras, LiDARs, lasers. I also communicated with customers regularly, and worked closely with the hardware team.
- Notable tasks:
    1) 3D reconstruction of a moving vehicle using line cameras and LiDAR + radar. Paper.
    2) Quality control for veneer using laser triangulation. 0.5mm accuracy from >1m away.
    3) Object tracking with multiple cameras through occlusions, 5cm accuracy from 6m away.
    4) OCR system on industrial packaging with majority voting reaching 99.9% accuracy.

Research assistant, LUT University, 04/2024 – 08/2024

- First authored a Springer LNCS conference paper on limited angle tomography using deep learning and image priors. paper

Freelance, Tmi Ilmari Vahteristo, 01/2024 - 04/2025

- Worked on reducing food waste at a restaurant by predicting customer volume based on menu.

OTHER TECH PROJECTS AND COMPETITIONS

Junction 2025, 11/2025

- Won Elevenlabs challenge for creating the best voice-based AI software. Demo

Research paper on numerical optimization for spherical codes, 02/2026 - 10/2026

- Developed a solver for spherical codes, beating the all-time record on 15% of cases. paper

LLM 20 questions, Kaggle, 06/2024 – 08/2024

- Ranked *7th out of 800* in creating an LLM based solution to play the "20 Questions" game.

Koodausta ja kisailua, Visma, 01/2023

- Won first place out of 20 competing teams. The task was to fix as many bugs as possible.

Google Foobar challenge, 05/2022 – 05/2023

- Nine difficult algorithmic problems. Got an interview at Google after completing.

IndySCC, CSC, 01/2022 – 11/2022

- Supercomputing competition, where we ranked 3rd from 10 universities across the world.

Reinforcement learning for a multiplayer card game, 10/2022 – 10/2023

- Trained an AI to play a card game with pure self-play, better than top humans. website, paper

EXTRA CURRICULAR

Project Manager, Lateksii, 01/2021 – 12/2022

- Organized events with our project team (4) + volunteers. Largest event 900 people.

Peacekeeper, UNIFIL, 11/2020 – 07/2021

- Squad co-lead, promoted part time to squad lead, and then team co-lead.

EDUCATION

- Graduated MSc (Tech), with 4.2 GPA, from Computer vision and pattern recognition, LUT university (08/2025) + exchange in Applied Math in Portugal, Lisbon, IST.
- Part-time 2nd MSc in Robotics and Automation engineering, Aalto University (estimated 9/2029)